10 Coordinate geometry: vectors and lines, a synthesis

Sheet 56 Tue 30 Mar. 2027 *Vectors and collinearity: consolidation*

1. Warm-up · Quick [GEO] $\vec{u}\begin{pmatrix}3\\-2\end{pmatrix}$ and $\vec{v}\begin{pmatrix}-1\\5\end{pmatrix}$. Calculate $\vec{u}+\vec{v}$ and $2\vec{u}$.

   Ans. $\begin{pmatrix}2\\3\end{pmatrix}$; $\begin{pmatrix}6\\-4\end{pmatrix}$.

2. Core · Method [GEO] Are the points $A(-1,2)$, $B(1,3)$ and $C(5,5)$ collinear?

   Ans. $\overrightarrow{AB}\begin{pmatrix}2\\1\end{pmatrix}$, $\overrightarrow{AC}\begin{pmatrix}6\\3\end{pmatrix}$: $2\times3-1\times6=0$: collinear.

3. Core + · Method [GEO] $A(-1,2)$, $B(3,1)$, $C(4,4)$. Find D such that $ABCD$ is a parallelogram.

   Ans. $\overrightarrow{AB}=\overrightarrow{DC}$: $D=A+\overrightarrow{BC}=(-1+1,2+3)=(0,5)$.

4. Review · Quick [PRO] In a class of 24 pupils, 18 have school lunch. What proportion is this, as a percentage?

   Ans. $\frac{18}{24}=0.75=75\%$.

5. Thread · Prove [LIT] Show that $(n+1)^2-(n-1)^2=4n$ for every integer n. What can be concluded about the difference of the squares of two consecutive odd numbers (n even)?

   Ans. $(n^2+2n+1)-(n^2-2n+1)=4n$. For $n=2k$ even, the difference of the squares of two consecutive odd numbers is $8k$: a multiple of 8.

   The two integers $n-1$ and $n+1$ are odd when n is even.

Sheet 57 Thu 1st Apr. 2027 *Determinant and equations of lines: consolidation*

1. Warm-up · Quick [GEO] What is the gradient of the line $3x+2y-6=0$?

   Ans. $y=-\frac{3}{2}x+3$: gradient $-\frac{3}{2}$.

2. Core · Method [GEO] Equation of the line (AB) with $A(1,2)$ and $B(4,8)$.

   Ans. Gradient $\frac{8-2}{4-1}=2$; $y=2x+p$, $2=2+p$: $y=2x$; or $2x-y=0$.

3. Core + · Method [GEO] Are the lines $2x-y+1=0$ and $4x-2y+5=0$ parallel, coincident or intersecting?

   Ans. Direction vectors $\begin{pmatrix}1\\2\end{pmatrix}$ and $\begin{pmatrix}2\\4\end{pmatrix}$ are collinear: parallel; $(0,1)$ satisfies the first but $4\times0-2+5=3\neq0$: strictly parallel.

4. Review · Quick [PUI] Calculate $10^{-2}\times5\times10^{4}$.

   Ans. $5\times10^{2}=500$.

5. Thread · Method [EQ] Solve in $\mathbb{R}$: $7-2(x-1)=3x-6$.

   Ans. $x=3$

   Pitfall: $-2(x-1)=-2x-2$ (sign error).

Sheet 58 Mon 5 Apr. 2027 *Plane geometry problems: choosing a method*

1. Warm-up · Quick [GEO] Distance AB for $A(0,0)$ and $B(5,12)$?

   Ans. $\sqrt{25+144}=13$.

2. Core · Method [DEM] $A(0,0)$, $B(4,2)$, $C(-2,4)$. What type of triangle is ABC?

   Ans. $AB^2=20$, $AC^2=20$, $BC^2=36+4=40=AB^2+AC^2$: right-angled isosceles at A.

3. Core + · Method [DEM] Give two methods to prove, using coordinates, that $ABCD$ is a parallelogram.

   Ans. (i) $\overrightarrow{AB}=\overrightarrow{DC}$; (ii) the diagonals $[AC]$ and $[BD]$ have the same midpoint.

4. Review · Reading [LEC] In a class of 30 pupils, a bar chart shows x pupils with the mark 12 and $2x$ pupils with the mark 14; the other marks account for 18 pupils. Find x.
   Ans. $x + 2x + 18 = 30$, so $3x = 12$ and $x = 4$.
5. Thread · Quick [SOL] A pyramid has a square base of side 6 cm and height 5 cm. Calculate its volume.
   Ans. $V = \frac{1}{3} \times 6^2 \times 5 = 60$ cm^3.

Sheet 59 Tue 6 Apr. 2027 *Situations from other subjects*

1. Warm-up · Quick [MOD] A moving object has position $x(t) = 3t + 2$ (in m, t in s). Where is it at $t = 4$?
   Ans. $x(4) = 14$ m.
2. Core · Method [MOD] A second object follows $x_2(t) = 5t - 4$. At what time and where do the objects meet?
   Ans. $3t + 2 = 5t - 4 \Leftrightarrow t = 3$ s; position $x = 11$ m.
3. Core + · Quick [GEO] A velocity vector has coordinates $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ (in m/s). What is its magnitude?
   Ans. $\sqrt{3^2 + 4^2} = 5$ m/s.
4. Review · Quick [CAL] Calculate $\frac{1}{2} - \frac{3}{5}$.
   Ans. $\frac{5}{10} - \frac{6}{10} = -\frac{1}{10}$.
5. Thread · Write-up [DEM] ABC is isosceles at A and H is the foot of the altitude from A. What can be stated about H?
   Ans. H is the midpoint of $[BC]$ (the altitude from the apex is also the perpendicular bisector of the base).

Sheet 60 Thu 8 Apr. 2027 *Communicating a proof: implication, converse, equivalence*

1. Warm-up · Quick [DEM] State the converse of: "if a quadrilateral is a rhombus, then its diagonals are perpendicular".
   Ans. "If a quadrilateral has perpendicular diagonals, then it is a rhombus".
2. Core · True/false [DEM] True or false? Justify. The previous converse is true.
   Ans. False. Counter-example: a kite (not a rhombus) has perpendicular diagonals.
3. Core + · Write-up [DEM] Is the equivalence "$x^2 = 4 \Leftrightarrow x = 2$" correct? Correct it.
   Ans. No: $x = -2$ also works. We have the implication $x = 2 \Rightarrow x^2 = 4$ and the equivalence $x^2 = 4 \Leftrightarrow (x = 2$ or $x = -2)$.
4. Review · Quick [CMP] Compare 2^{10} and 10^3.
   Ans. $2^{10} = 1{,}024 > 1{,}000 = 10^3$.
5. Thread · Quick [REP] Between which consecutive integers does $\frac{7}{4}$ lie on a number line? Give its decimal value.
   Ans. Between 1 and 2; 1.75.

Sheet 61 Mon 26 Apr. 2027 *Class discussion: comparing methods*

1. Warm-up · Quick [GEO] Distance between $A(-2, 1)$ and $B(1, 5)$?
   Ans. $\sqrt{3^2 + 4^2} = 5$.
2. Core · Method [GEO] Give three methods to prove that A, B, C are collinear.
   Ans. (i) $\det\left(\overrightarrow{AB}, \overrightarrow{AC}\right) = 0$; (ii) the equation of (AB) is satisfied by C; (iii) $AB + BC = AC$: equality of distances, equivalent to $B \in [AC]$.

3. Core + · Reflection [GEO] To prove that two lines given by $y = m_1x + p_1$ and $y = m_2x + p_2$ are parallel, which method is quickest?

Ans. Compare the gradients: $m_1 = m_2$; strictly parallel if in addition $p_1 \neq p_2$.

4. Review · Factorise [DEV] Factorise $6x + 9$.

Ans. $3(2x + 3)$.

5. Thread · Method [DIV] Write 360 as a product of prime factors. How many divisors does it have?

Ans. $360 = 2^3 \times 3^2 \times 5$; $(3+1)(2+1)(1+1) = 24$ divisors.